

Efficient Subfield Attacks on Overstretched NTRU

Abstract—The subfield attack maps an NTRU instance down to a subfield, reduces a smaller lattice there and lifts the solution back. Two obstacles have kept this family without a predictive cost model. First, the shortest vector of the descended lattice, which decides whether reduction recovers the key, has never been determined beyond the bound given by the descended secret. Second, concrete estimates apply the Gaussian heuristic to sections of the planted dense sublattice, and that heuristic fails when the planted lattice behaves like a rotation of the integer lattice $\mathbb{Z}^n$, with n the ring degree, while its validity range has remained unknown.

In this work we settle both questions and build a faster attack on them. We give the exact law of the covolume of the descended dense sublattice, which involves a constant c_r depending only on the relative degree r , and predict the shortest vector of the descended lattice with a mean error of 4%. We then obtain the validity range of the Gaussian heuristic through an explicit excess exponent, show that the crossover is gradual, and cut the prediction error from 19% to 9%. Our attack selects the relative degree from an exact cost model, uses a threshold provably a factor Δ_r weaker than the previous one, and adds a module reduction shortening the recovered vector by up to $6.2\times$. We reimplement all earlier attacks and measure gains up to $55\times$ in time and $4.7\times$ in memory against the descents, and three orders of magnitude against full-field reduction.

I. INTRODUCTION

The NTRU problem of Hoffstein, Pipher and Silverman [1] asks to recover short ring elements f, g from $h = gf^{-1}$ modulo an integer q , and it underlies encryption, signatures and homomorphic encryption [2], [3], [4], [5] as well as candidate multilinear maps [6]. Its hardness is measured by lattice reduction [7], [8] on the NTRU lattice Λ_q^h , of dimension $2n$ with n the ring degree [9]. Beyond a threshold called the fatigue point the security of NTRU falls noticeably below that of Ring-LWE [10] at the same parameters, a regime known as overstretched [11], [12], [13], [14]. Two families operate there. Subfield attacks [11], [12], [13] send the instance into a subfield L of relative degree r through the relative norm or trace, reduce a lattice of dimension $2n/r$ and lift the outcome, while geometric attacks [13], [14] reduce the full lattice and exploit the dense rank- n sublattice it contains [15]. The choice between the two families, and the choice of r , is currently made without a predictive model.

Problem 1: the minimum of the descended lattice. Write $f' = N_{K/L}(f)$, $g' = N_{K/L}(g)$ and $\Lambda_q^{h'}$ for the descended secrets and lattice. Albrecht, Bai and Ducas [11] name this as the obstruction to concrete predictions for subfield attacks, because the shortest vector of $\Lambda_q^{h'}$ is not known and all that is available is the upper bound $\|(f', g')\|$, which their own experiments show to be loose; they add that reduction here may

be governed by many short vectors instead of a few. Kirchner and Fouque [13] point to the same gap when they observe that the two earlier subfield attacks came without a precise analysis of their running time, and the same absence is echoed in [14].

Problem 2: the Gaussian heuristic on sections of a dense sublattice. The estimator of Ducas and van Woerden [14] predicts the block size at which reduction discovers the dense sublattice $\mathcal{D}$. It needs the first minimum of $\mathcal{D} \cap V$, with V the span of the leading part of the current basis, and takes it from the Gaussian heuristic. The authors point out that the heuristic is false when the section is all of $\mathcal{D}$, because $\mathcal{D}$ is close to a rotation of $\mathbb{Z}^n$, that it holds when the section is small, and they leave the location and the nature of the crossover as an open problem. How far the geometry of NTRU lattices helps reduction is also raised in [16], and the reliability of such predictions is questioned in [17], [18].

Contribution 1 (Section III) solves Problem 1. The descended lattice carries a dense sublattice $\mathcal{D}' = (f', g')\mathcal{O}_L$ of rank $n' = n/r$ whose normalized covolume obeys an exact law with logarithm $\frac{1}{2}(r \ln(n\sigma^2) + \mu_r)$, where σ^2 is the coefficient variance of the secret and $\mu_r = -r\gamma + \mathbb{E}[\ln(2 \cosh(D_r/2))]$ with γ the Euler constant and D_r a sum of r independent standard logistic variables. We prove a two-sided bound on $\lambda_1(\Lambda_q^{h'})$, give a model that experiments confirm with a mean error of 4%, and improve the recovery threshold of [11] by a provable factor $\Delta_{r,n'}$ reaching 12.4 at $r = 16$.

Contribution 2 (Section IV) solves Problem 2. We introduce the excess factor of a lattice at a radius, the ratio between its true point count and the count of the Gaussian heuristic, and show that the minimum of a section of rank k is the heuristic value divided by the excess factor to the power $1/k$. For a dense sublattice close to a rotation of $\mathbb{Z}^n$ we evaluate the excess exponent by a saddle-point argument agreeing with exact integer point counts to within $3 \cdot 10^{-3}$. The criterion locates the crossover, shows that it is gradual and not abrupt, and lowers the mean prediction error from 19.2% to 8.6%.

Contribution 3 (Sections V, VI) is a faster subfield attack. It picks the relative degree and the ambient dimension from a cost model built on the first two contributions, descends through the automorphism tower, reduces with the relaxed threshold, and adds a module reduction in dimension n' returning the shortest element of $\mathcal{D}'$. That step lowers the smallest modulus at which a relative degree is usable by up to 12.1 bits, which lets the attack work in half the dimension of the earlier ones. We reimplement the full-field reduction and the three earlier descents [11], [12], [13] and compare them on the same instances and machine; Table I summarizes the outcome and

TABLE I: Comparison with the attacks in the overstretched regime, with d the reduction dimension, β the halting block size, r the relative degree, $n' = n/r$, ℓ the ambient coordinates kept, $\Delta_{r,n'}$ the shape defect of Definition 6, γ_k the Gaussian-heuristic constant, and $f' = N_{K/L}(f)$, $g' = N_{K/L}(g)$, $\tilde{g}' = \text{Tr}_{K/L}(gF)$ with $F = \prod_{\psi \neq 1} \psi(f)$ the descended secrets. Time counts bit operations and space bits, with T and S the sieve costs of Section II and $O(n \log r \log q)$ for the descent in every row. The planted volume is the value of $\text{covol}(\mathcal{D}')$ the method uses, the section minimum the value it gives to the first minimum of a section, the threshold the length below which a reduced vector is accepted, and the last column the largest gain in time and in memory measured against the row.

Method	d	Time	Space	Planted volume	Section minimum	Threshold	Model r	Measured gain
Full-field reduction [13], [14]	$2n$	$n^4 \log q + nT(\beta)$	$\max\{n^2 \log q, S(\beta)\}$	exact for $r = 1$	Gaussian heuristic	$\gamma_{2n} \sqrt{q}$	$r = 1$ only	$3.5 \cdot 10^3 \times / 34 \times$
Norm descent [11]	$2n'$	$n'^4 \log q + n'T(\beta)$	$\max\{n'^2 \log q, S(\beta)\}$	Hadamard	not used	$q/\ (f', g')\ $	no	$55 \times / 4.6 \times$
Trace descent [12]	$2n'$	$n'^4 \log q + n'T(\beta)$	$\max\{n'^2 \log q, S(\beta)\}$	Hadamard	not used	$q/\ (f', g')\ $	no	$47 \times / 4.7 \times$
Subring projection [13]	$\ell + n'$	$d^4 \log q + dT(\beta)$	$\max\{d^2 \log q, S(\beta)\}$	Hadamard	Gaussian heuristic	Pataki-Tural	partly	$5.3 \times / 3.5 \times$
This work	$\min(r, \ell)$	$d^4 \log q + dT(\beta) + n'^4 \log q$	$\max\{d^2 \log q, S(\beta)\}$	exact, Theorem 5	corrected, Theorem 11	$q/\text{covol}(\mathcal{D}')^{1/n'}$	yes	baseline

all proofs are in the full version.

II. PRELIMINARIES

Lattices. A lattice Λ of rank k has covolume $\text{covol}(\Lambda)$, the square root of the Gram determinant of a basis, and first minimum $\lambda_1(\Lambda)$. With $V_k = \pi^{k/2}/\Gamma(k/2 + 1)$ the volume of the unit ball and $\gamma_k := \max\{1, V_k^{-1/k}\} \sim \sqrt{k/(2\pi e)}$, the Gaussian heuristic equates the number of lattice points in a ball with the ratio of the ball volume to the covolume and predicts $\lambda_1(\Lambda) \approx \text{gh}(\Lambda) := \gamma_k \text{covol}(\Lambda)^{1/k}$, while Minkowski's theorem gives $\lambda_1(\Lambda) \leq 2\gamma_k \text{covol}(\Lambda)^{1/k}$. Block reduction with block size β [7], [8] returns a basis whose Gram-Schmidt profile follows a geometric series of ratio δ_β^{-2} , with δ_β the root-Hermite factor of the simulator of [8] used in [19], [20]. One call to a shortest vector oracle in dimension β costs $2^{0.292\beta+16.4}$ operations and $2^{0.2075\beta+16.4}$ vectors of memory [21], [22], and a quantum sieve lowers the time exponent to 0.257β [23], which shifts all costs by one common factor.

Cyclotomic towers. Let n be a power of two, $K = \mathbb{Q}[X]/(X^n + 1)$ and $\mathcal{O}_K = \mathbb{Z}[X]/(X^n + 1)$, a rescaled copy of $\mathbb{Z}^n$ in the coefficient embedding. For a power of two $r \mid n$ the subfield $L = \mathbb{Q}[Y]/(Y^{n'} + 1)$ with $n' = n/r$ embeds through $Y \mapsto X^r$, and $\mathcal{O}_L = \mathbb{Z}[X^r]/(X^n + 1)$. The group $H = \text{Gal}(K/L)$ has order r , the relative norm is $N_{K/L}(a) = \prod_{\psi \in H} \psi(a)$ and the relative trace is $\text{Tr}_{K/L}(a) = \sum_{\psi \in H} \psi(a)$; both obtained by repeating $a \mapsto a(X)a(-X)$, respectively $a \mapsto a(X) + a(-X)$. We write $\iota : L \rightarrow K$ for the inclusion and σ' , σ for embeddings of L and of K . For $n' \geq 2$ the r embeddings above a fixed σ' are pairwise non-conjugate, so the $n'/2$ conjugate classes of L lift to disjoint groups of r classes of K .

The NTRU problem. Fix q and a distribution on $\mathcal{O}_K$ with independent centered coefficients of variance σ^2 , so $\sigma^2 = 2/3$ for ternary secrets. Sample f, g with f invertible modulo q and publish $h = [gf^{-1}]_q$. As in [11], solving $\text{NTRU}(\mathcal{O}_K, q, \sigma, \tau)$ means finding a nonzero vector of length at most τ in $\Lambda_q^h = \{(x, y) \in \mathcal{O}_K^2 : y \equiv xh \pmod{q}\}$, of dimension $2n$ and volume q^n . Our target is $\tau = \gamma_{2n} \sqrt{q}$, the length below which a random lattice of that volume holds no vector, so any solution distinguishes h from uniform and breaks the schemes of [4], [5], whose keys are too narrow to make h

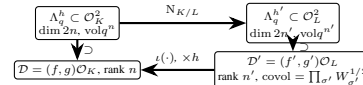


Fig. 1: Objects used by the attack. The public lattice carries the rank- n dense sublattice $\mathcal{D}$ spanned by the rotations of the key, the relative norm sends the instance to L , where $\mathcal{D}'$ has rank n' and the covolume Lemma 3 gives exactly, and ι followed by one product with h modulo q returns a multiple of the key.

statistically uniform [24]. The lattice contains the rank- n dense sublattice $\mathcal{D} = \{(vf, vg) : v \in \mathcal{O}_K\}$, whose discovery drives the overstretched regime [13], [14].

Definition 1 (descended instance). For $r \mid n$ set $f' = N_{K/L}(f)$, $g' = N_{K/L}(g)$ and $h' = N_{K/L}(h)$, so $g' \equiv f'h' \pmod{q}$. The descended lattice is $\Lambda_q^{h'} = \{(x, y) \in \mathcal{O}_L^2 : y \equiv xh' \pmod{q}\}$, of dimension $2n'$ and volume $q^{n'}$, and the descended dense sublattice is $\mathcal{D}' = (f', g')\mathcal{O}_L$, of rank n' .

The attack of [11] reduces $\Lambda_q^{h'}$, checks that the returned vector lies in $\mathcal{D}'$ and lifts it, [12] uses the trace in place of the norm, and [13] keeps $\ell \leq n$ ambient coordinates and descends only the second block, of dimension $\ell + n'$ and volume q^ℓ . All three reduce a lattice carrying a planted rank- n' sublattice, the object we analyze.

III. THE MINIMUM OF THE DESCENDED LATTICE

Throughout, $r \geq 2$, $n' = n/r$, and f, g are as in Section II.

Definition 2 (shape weights). For each embedding σ' of L the shape weight is $W_{\sigma'} := |\sigma'(f')|^2 + |\sigma'(g')|^2$.

The weights are the eigenvalues of the quadratic form carried by $\mathcal{D}'$.

Lemma 3. The map $v \mapsto (vf', vg')$ sends $\mathcal{O}_L$ onto $\mathcal{D}'$ and its Gram matrix in the coefficient basis has eigenvalues $W_{\sigma'}$, so $\text{covol}(\mathcal{D}') = \prod_{\sigma'} W_{\sigma'}^{1/2}$ and $\|(f', g')\|^2 = \frac{1}{n'} \sum_{\sigma'} W_{\sigma'}$.

Proof. In the basis $1, Y, \dots, Y^{n'-1}$ multiplication by $a \in \mathcal{O}_L$ is the negacyclic matrix M_a , and negacyclic matrices are simultaneously diagonalized by the unitary matrix U of evaluations at the primitive $2n'$ -th roots of unity, $M_a = U \text{diag}(\sigma'(a))_{\sigma'} U^*$. The Gram matrix of the map is

$M_{f'}^\top M_{f'} + M_{g'}^\top M_{g'} = U \operatorname{diag}(W_{\sigma'})_{\sigma'} U^*$, whose eigenvalues are the $W_{\sigma'}$ and whose determinant has square root $\prod_{\sigma'} W_{\sigma'}^{1/2}$. The second identity is Parseval for the negacyclic transform, $\sum_{\sigma'} |\sigma'(a)|^2 = n' \|a\|^2$, applied to f' and g' . $\square$

Lemma 4 bounds from both sides the quantity left undetermined in [11].

Lemma 4. *Any nonzero $w \in \mathcal{D}'$ satisfies $\|w\| \geq \operatorname{covol}(\mathcal{D}')^{1/n'}$, hence*

$$\begin{aligned} \operatorname{covol}(\mathcal{D}')^{1/n'} &\leq \lambda_1(\mathcal{D}') \\ &\leq \min\{\|(f', g')\|, 2\gamma_{n'} \operatorname{covol}(\mathcal{D}')^{1/n'}\}, \end{aligned}$$

and $\lambda_1(\Lambda_q^{h'}) = \min\{\lambda_1(\mathcal{D}'), \lambda_1(\Lambda_q^{h'} \setminus \mathcal{D}')\}$, the second term being $\gamma_{2n'} \sqrt{q}$ under the usual heuristic for random q -ary lattices.

Proof. Write $w = (vf', vg')$ with $v \neq 0$. By Lemma 3 and Parseval, $\|w\|^2 = \frac{1}{n'} \sum_{\sigma'} |\sigma'(v)|^2 W_{\sigma'}$, and the inequality between the arithmetic and the geometric mean gives $\|w\|^2 \geq (\prod_{\sigma'} |\sigma'(v)|^2 W_{\sigma'})^{1/n'} = |N_{L/\mathbb{Q}}(v)|^{2/n'} \operatorname{covol}(\mathcal{D}')^{2/n'}$. A nonzero algebraic integer has $|N_{L/\mathbb{Q}}(v)| \geq 1$, which proves the lower bound. The first upper bound is $v = 1$ and the second is Minkowski's theorem. The last claim holds because $\mathcal{D}' \subseteq \Lambda_q^{h'}$ partitions the nonzero vectors. $\square$

Theorem 5 gives the exact law of the quantity that controls both sides.

Theorem 5. *Let f, g have independent Gaussian coefficients of variance σ^2 and let D_r be a sum of r independent standard logistic variables. Then $\mathbb{E}[\ln \operatorname{covol}(\mathcal{D}')] = \frac{n'}{2}(r \ln(n\sigma^2) + \mu_r)$ with*

$$\mu_r = -r\gamma + \mathbb{E}[\ln(2 \cosh(D_r/2))],$$

γ the Euler constant; the variance of $\ln \operatorname{covol}(\mathcal{D}')$ is $O(n'r)$, $\mu_1 = 1 - \gamma$, and $\mu_r = -r\gamma + \sqrt{\pi r/6} + O(1)$ for large r .

Proof sketch. For a power-of-two cyclotomic field the values $\sigma_j(f)$ at one representative of each conjugate class are independent complex Gaussians of variance $n\sigma^2$, so $|\sigma_j(f)|^2 = n\sigma^2 a_j$ with a_j independent standard exponentials. The r embeddings above one σ' are pairwise non-conjugate, so $|\sigma'(f')|^2 = (n\sigma^2)^r \prod_{i \leq r} a_i$ and likewise for g' with independent b_i , whence $W_{\sigma'} = (n\sigma^2)^r \Psi$ with $\Psi = \prod a_i + \prod b_i$, independent across the $n'/2$ classes. Writing $\ln(A+B) = \frac{1}{2} \ln(AB) + \ln(2 \cosh(D/2))$ with $D = \ln A - \ln B$, using $\mathbb{E}[\ln a] = -\gamma$ and the fact that the difference of the logarithms of two independent standard exponentials is standard logistic gives $\mathbb{E}[\ln \Psi] = \mu_r$; Lemma 3 converts this into the stated expectation, and the variance follows because $\ln \operatorname{covol}$ is a sum of $n'/2$ independent terms of variance $O(r)$. The asymptotic form comes from the central limit theorem for D_r , whose variance is $r\pi^2/3$. The full argument is in the appendix of the full version. $\square$

Write $c_r = e^{\mu_r}$ and $\rho_r := \operatorname{covol}(\mathcal{D}')^{1/n'} = ((n\sigma^2)^r c_r)^{1/2}$, and let $\nu_{r,m} = \mathbb{E}[\ln(\frac{1}{m} \sum_{i \leq m} \Psi^{(i)})]$ for independent copies

of $\Psi = \prod_{i \leq r} a_i + \prod_{i \leq r} b_i$, with a_i, b_i independent standard exponential variables and $m = n'/2$, so that $\mathbb{E}[\ln \|(f', g')\|] = \frac{1}{2}(r \ln(n\sigma^2) + \nu_{r,m})$.

Definition 6 (shape defect). $\Delta_{r,n'} := \exp(\frac{1}{2}(\nu_{r,n'/2} - \mu_r))$, so that $\|(f', g')\| = \Delta_{r,n'} \rho_r$ in log expectation.

The defect grows quickly with the relative degree.

Corollary 7. $\Delta_{r,n'} \geq 1$, and for fixed r and growing n' it tends to $\sqrt{2/c_r} = \sqrt{2} e^{r\gamma/2 - \sqrt{\pi r/6}/2 + O(1)}$, which grows exponentially in r ; for $r = 1$ the limit is $\sqrt{2} e^{(\gamma-1)/2} = 1.1447$.

Proof. The inequality $\nu \geq \mu$ is the inequality between the arithmetic and the geometric mean applied inside the expectation. For fixed r and $m \rightarrow \infty$ the empirical mean of the $\Psi^{(i)}$ converges to $\mathbb{E}[\Psi] = 2$ in L^1 and almost surely, so $\nu_{r,m} \rightarrow \ln 2$ and $\Delta \rightarrow \sqrt{2/c_r}$; substituting $c_r = e^{\mu_r}$ and the asymptotic form of Theorem 5 gives the displayed expression, and $r = 1$ gives $\sqrt{2/e^{1-\gamma}}$. $\square$

A twisted copy of $\mathbb{Z}^{n'}$ obeys the Gaussian heuristic once its weights are spread enough, which with Lemma 4 gives the model

$$\lambda_1(\mathcal{D}') = \min\{\Delta_{r,n'}, \gamma_{n'}\} \cdot \rho_r, \quad (1)$$

interpolating between the two upper bounds of Lemma 4, since for small r the weights are flat and the minimum is attained at $v = 1$, while for large r their spread makes the heuristic value binding; Section VI confirms it with a mean error of 4%. The exact covolume also sharpens the recovery threshold, which [11] derive from $\operatorname{covol}(\mathcal{D}') \leq \|(f', g')\|^{n'}$.

Theorem 8. *Assume $\langle f' \rangle$ and $\langle g' \rangle$ are coprime. If $(x', y') \in \Lambda_q^{h'}$ satisfies $\|(x', y')\| < q/\rho_r$ then $(x', y') \in \mathcal{D}'$. The admissible length exceeds the one of [11] by the factor $\Delta_{r,n'}$.*

Proof sketch. Coprimality yields (F', G') with $f'G' - g'F' = q$, and the $\mathcal{O}_L$ -module generated by (f', g') and (F', G') contains $q\mathcal{O}_L^2$ and $(1, h')$ while having the same covolume as $\Lambda_q^{h'}$, hence equals it. Multiplication by X is an isometry of the coefficient embedding, so the subspace of vectors perpendicular to $\operatorname{span}(\mathcal{D}')$ is stable under $\mathcal{O}_L$ and the projection $\Lambda^\perp$ of $(F', G')\mathcal{O}_L$ onto it is again a rank-one $\mathcal{O}_L$ -module with $\operatorname{covol}(\mathcal{D}') \operatorname{covol}(\Lambda^\perp) = q^{n'}$. The lower bound of Lemma 4, whose proof uses only the rank-one module structure, gives $\lambda_1(\Lambda^\perp) \geq q/\rho_r$, and a vector shorter than that has zero component outside $\mathcal{D}'$. Using $\operatorname{covol}(\mathcal{D}') \leq \|(f', g')\|^{n'}$ instead loses exactly $\Delta_{r,n'}$ by Definition 6. The full argument is in the appendix of the full version. $\square$

IV. VALIDITY OF THE GAUSSIAN HEURISTIC ON SECTIONS

Estimates for overstretched NTRU predict the block size at which reduction discovers $\mathcal{D}$ by comparing the projection of a short vector of a section of $\mathcal{D}$ against the Gram-Schmidt profile [13], [14]. With $B = (b_0, \dots, b_{d-1})$ the current basis of a lattice Λ of dimension d carrying a dense sublattice $\mathcal{D}$ of rank m , the object needed is the section $\mathcal{D}_s := \mathcal{D} \cap$

$\text{span}(b_0, \dots, b_{s-1})$ of rank $k = s - (d - m)$, for which [14] sets $\lambda_1(\mathcal{D}_s) = \text{gh}(\mathcal{D}_s)$, accurate for small k and false by a factor $\Theta(\sqrt{m})$ at $k = m$.

Definition 9 (excess factor). For a lattice Λ of rank m and a radius $R > 0$, let $N_\Lambda(R)$ be the number of nonzero points of length at most R . The excess factor and its exponent at normalized radius c are

$$\varrho_\Lambda(R) := N_\Lambda(R) \text{covol}(\Lambda) / (V_m R^m),$$

$$\Xi_\Lambda(c) := \frac{1}{m} \ln \varrho_\Lambda(c \sqrt{m} \text{covol}(\Lambda)^{1/m}).$$

Sections are cut by subspaces unrelated to the arithmetic of $\mathcal{D}$, which motivates the only heuristic of this section.

Heuristic 10. For a section $\mathcal{D}_s$ of rank k , the expected number of nonzero points of length at most R is $\varrho_{\mathcal{D}}(R) \cdot V_k R^k / \text{covol}(\mathcal{D}_s)$.

It states that a fixed point of $\mathcal{D}$ of length R lies in the span of the leading basis vectors with a probability depending on R only through the volume ratio, the calibration that makes the two counts agree whenever $\mathcal{D}$ itself follows the heuristic.

The excess factor fixes the minimum of every section.

Theorem 11. Under Heuristic 10 the first minimum of the section of rank k satisfies

$$\lambda_1(\mathcal{D}_s) = \max \left\{ \lambda_1(\mathcal{D}), \text{gh}(\mathcal{D}_s) e^{-\frac{m}{k} \Xi_{\mathcal{D}}(u)} \right\}, \quad (2)$$

$$u = \lambda_1(\mathcal{D}_s) / (\sqrt{m} \text{covol}(\mathcal{D})^{1/m}).$$

If $\mathcal{D}$ is a scaled rotation of $\mathbb{Z}^m$ then $\Xi_{\mathcal{D}}$ is smooth and decreasing, with $\Xi_{\mathcal{D}}(u) = -\ln(u\sqrt{2\pi e}) + o(1)$ as $u \rightarrow 0$ and $\Xi_{\mathcal{D}}(u) = 2e^{-2\pi^2 u^2} (1 + o(1))$ as u grows. The Gaussian heuristic then holds for $\mathcal{D}_s$ within a factor $1 + \varepsilon$ exactly when

$$u \geq u^*(k, \varepsilon) := (2\pi^2)^{-1/2} \ln^{1/2}(2m/(k \ln(1 + \varepsilon))), \quad (3)$$

and the ratio $\lambda_1(\mathcal{D}_s) / \text{gh}(\mathcal{D}_s)$ moves from e^{-1} to $e^{-1/100}$ over an interval of u^2 of length $\ln(100)/(2\pi^2) = 0.233$, of the same order as $(u^*)^2$, so the crossover is gradual and not abrupt.

Proof sketch. Setting the expected count of Heuristic 10 to one gives $\lambda_1(\mathcal{D}_s) = \text{gh}(\mathcal{D}_s) \varrho_{\mathcal{D}}(\lambda_1(\mathcal{D}_s))^{-1/k}$, which is (2) once the inclusion $\mathcal{D}_s \subseteq \mathcal{D}$ is imposed as a lower bound. For $\Lambda = \mathbb{Z}^m$ the count in a ball of radius $u\sqrt{m}$ is evaluated by exponential tilting: with $\theta_3(t) = \sum_{z \in \mathbb{Z}} e^{-\pi t z^2}$ and t solving $m \mathbb{E}_t[z^2] = u^2 m$, the count is $\theta_3(t)^m e^{\pi t u^2 m}$ divided by the fluctuation factor of a local limit theorem, and subtracting the heuristic exponent $m \ln(u\sqrt{2\pi e})$ gives Ξ . Smoothness and the decrease follow from those of θ_3 with the bound $\mathbb{E}_t[z^2] \leq 1/(2\pi t)$ for a discrete Gaussian; the two limits follow from $\theta_3(t) = 1 + 2e^{-\pi t} + O(e^{-4\pi t})$ and from $\theta_3(t) = t^{-1/2} \theta_3(1/t)$. Inequality (3) is $\frac{m}{k} \Xi \leq \ln(1 + \varepsilon)$ with the large- u form of Ξ , and the width follows by solving it at the two stated levels. The full argument is in the appendix of the full version. $\square$

Against the exact number of integer points in balls of squared radius up to 4 in dimensions 32, 64 and 128, the evaluation of Ξ has error below $2.6 \cdot 10^{-3}$.

The criterion says where the heuristic may be used.

Corollary 12. The Gaussian heuristic is accurate for sections of rank k as long as $\text{gh}(\mathcal{D}_s) \geq \sqrt{m} \text{covol}(\mathcal{D})^{1/m} u^*(k, \varepsilon)$, a threshold growing like $\sqrt{\ln(m/k)}$, and it fails for every k with $\text{gh}(\mathcal{D}_s) < \lambda_1(\mathcal{D})$, where it underestimates the minimum by at least the factor $\lambda_1(\mathcal{D}) / \text{gh}(\mathcal{D}_s)$.

Proof. The first part is (3) rewritten through $u = \lambda_1(\mathcal{D}_s) / (\sqrt{m} \text{covol}(\mathcal{D})^{1/m})$ together with $\lambda_1(\mathcal{D}_s) \approx \text{gh}(\mathcal{D}_s)$ in the regime where the heuristic is accurate. The second follows from $\mathcal{D}_s \subseteq \mathcal{D}$, which forces $\lambda_1(\mathcal{D}_s) \geq \lambda_1(\mathcal{D})$, so a heuristic value below $\lambda_1(\mathcal{D})$ is an underestimate by at least the stated ratio. $\square$

Substituting (2) into the dense sublattice discovery criterion of [14] gives the corrected estimator used in Section V; since $\varrho \geq 1$ it can only lower the predicted block size.

V. THE ATTACK AND ITS COMPLEXITY

Our attack takes the public key and the target τ and returns a nonzero vector of Λ_q^h of length at most τ that is an $\mathcal{O}_K$ -multiple of the key. Its predictor evaluates, for each power of two $r \mid n$ and each ℓ in a small grid, the normalized covolume of Theorem 5, the minimum (1), the block size at which the corrected criterion of Section IV fires, and the resulting time and memory. Every length is evaluated at the mean plus two and a half standard deviations of $\ln \text{covol}(\mathcal{D}')$, an offset entering the four earlier rules as well, so no method is favored. A configuration is admissible when the planted sublattice is denser than the ambient lattice by at least a quarter of the slack the Pataki–Tural bound leaves on the profile one reduction pass produces, when the vector it plants can be recognized through Theorem 8, and when the lifted vector is predicted to be at most τ ; the last condition is where the module reduction pays off, since it lowers the predicted lifted length by $\max\{1, \Delta_{r,n'}^{\text{lift}} / \gamma_{n'}\}$ and so admits larger r at the same q . Figure 1 shows the objects involved.

Algorithm 1: Subfield attack with descended dense sublattice discovery

Input: public key $h \in \mathcal{O}_K / q$, parameters n, q, σ^2 , target τ
Output: $(x, y) \in \Lambda_q^h$ with $0 < \|(x, y)\| \leq \tau$

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1  $(r, \ell, \beta_{\max}) \leftarrow \text{Predict}(n, q, \sigma^2, \tau); \quad h' \leftarrow h$ 
2 for  $i = 1$  to  $\log_2 r$  do  $h' \leftarrow$  even part of  $h'(X)h'(-X) \bmod q$ , in  $X^2$ 
3  $\rho \leftarrow ((n\sigma^2)^r c_r)^{1/2}; \quad B \leftarrow$  basis of  $\Lambda(r, \ell); \quad \beta \leftarrow 2$ 
4 while  $\beta \leq \beta_{\max}$  do
5    $B \leftarrow \text{Reduce}(B, \beta); \quad Y \leftarrow \{\mathcal{O}_L\text{-part of the short rows of } B\}$ 
6   foreach  $y \in Y$  do
7      $x \leftarrow \iota(y); \quad y^\dagger \leftarrow [xh]_q$  centered
8     if  $0 < \|(x, y^\dagger)\| \leq \tau$  then return  $(x, y^\dagger)$ 
9   end
10   $S \leftarrow \{y \in Y : \|(y, [yh']_q)\| < q/\rho\}$ 
11   $M \leftarrow \mathcal{O}_L\text{-module generated by the two shortest pairs from } S$ 
12   $y^* \leftarrow \mathcal{O}_L\text{-part of a shortest vector of } \text{Reduce}(M, 20);$  lift and test it as
    above
13   $\beta \leftarrow \beta + 1$ 
14 end
```

Algorithm 1 takes the key h , the parameters n, q, σ^2 and the target τ , and calls the predictor, which returns the relative degree r , the ambient size ℓ and a cap on the block size. It

descends the key by squaring the tower $\log_2 r$ times, each step multiplying the current element by its image under $X \mapsto -X$ and reading the product in X^2 , then sets the threshold ρ and builds the basis. The main loop reduces at the current block size and collects the $\mathcal{O}_L$ -parts of the short rows; each is lifted by embedding it into $\mathcal{O}_K$ and recomputing the second coordinate modulo q , and the loop returns once a lifted pair meets the target. Otherwise the rows shorter than q/ρ , which Theorem 8 places in $\mathcal{D}'$, are paired with their companions $[yh']_q$ so that each pair generates a rank- n' module; the module of the two shortest pairs is reduced, its shortest element is lifted and tested, and the block size then grows by one.

The lift returns a multiple of the secret key.

Lemma 13. *If $y = vf'$ for some $v \in \mathcal{O}_L$ and $\|\iota(v)F(f, g)\| < q/2$ with $F = \prod_{\psi \in H \setminus \{1\}} \psi(f)$, then the lift returns $\iota(v)F \cdot (f, g)$, an $\mathcal{O}_K$ -multiple of the secret key lying in Λ_q^h .*

Proof. Since $\iota(N_{K/L}(f)) = fF$ we have $x = \iota(y) = \iota(v)fF$, hence $xh = \iota(v)fF \cdot gf^{-1} = \iota(v)Fg$ in K , an identity that also holds modulo q . The centered representative of $[xh]_q$ therefore equals $\iota(v)Fg$ once the latter has infinity norm below $q/2$, which the hypothesis gives. The returned pair is $(\iota(v)Ff, \iota(v)Fg) = \iota(v)F(f, g)$, and it lies in Λ_q^h because (f, g) does and the lattice is an $\mathcal{O}_K$ -module. $\square$

The cost follows from the descent and the reduction.

Theorem 14. *Let d be the reduction dimension and β the block size at which Algorithm 1 halts. Its running time is $O(n \log r \log q \log \log q) + O(d^4 \log q) + O(d\beta^{2^{0.292\beta+16.4}})$ bit operations and its memory is $O(\max\{d^2 \log q, \beta^{2^{0.2075\beta+16.4}}\})$ bits, the module reduction adding $O(n^4 \log q)$ bit operations and $O(n^2 \log q)$ bits.*

Proof sketch. Level i of the descent multiplies two polynomials of degree $n/2^i$ with coefficients of $\log q$ bits at cost $O((n/2^i) \log(n/2^i) \log q \log \log q)$ with a fast product, and the $\log_2 r$ levels sum to the first term; the lift is one further product of the same shape. The reduction performs one call to the L^2 algorithm, of cost $O(d^4 \log q)$ on a d -dimensional basis with entries of $\log q$ bits [25], followed by at most β progressive tours, each making $O(d)$ calls to a shortest vector oracle in dimension β at the cost and memory of Section II. The basis occupies $O(d^2 \log q)$ bits, and the module reduction runs on a rank- n' lattice in $\mathbb{Z}^{2n'}$ with entries bounded by q . The full argument is in the appendix of the full version. $\square$

The last two columns of Table I collect these expressions for all five attacks. All rows share a descent, one call to the L^2 algorithm and a sieving loop, so they differ only through d and β ; the family our predictor searches contains the lattice of every other row, and its β comes from the corrected criterion of Section IV and the relaxed threshold of Theorem 8, both of which can only lower it, so the last row dominates in time and in memory at once.

VI. EXPERIMENTAL EVALUATION

All measurements use one core of an Apple M2 with 16 GB of memory, Python 3.12 and fpylll 0.6.4 over fplll 5.5.0 [26], [27]. Secrets are ternary, the modulus is the smallest prime above a power of two, and each attack runs in its own process whose peak resident memory is measured once the instance and the model constants are ready, so the memory shown is that of the reduction. Each configuration runs on three instances at $n \leq 256$ and two at $n = 512$, and we give medians. All five attacks are our own implementations, and success means the returned pair lies in Λ_q^h , is shorter than τ and is an $\mathcal{O}_K$ -multiple of the key, verified over the integers.

Descended covolume and minimum. The first two panels of Figure 2 compare Theorem 5 and the model (1) against 15 and 12 configurations with $n \leq 1024$ and $r \leq 16$. The covolume law is reproduced to within 0.024 in a logarithmic quantity ranging from 2.8 to 49.1, and the shape defect to within 7.2%. The measured ratio $\lambda_1(\mathcal{D}')/\text{covol}(\mathcal{D}')^{1/n'}$ follows $\min\{\Delta_{r,n'}, \gamma_{n'}\}$ with a mean error of 4.1%, and with an error between -6.5% and $+6.6\%$ in the eleven configurations where $n' \geq 32$; at $n' = 16$ the measured minimum exceeds the model by 17.1%, because the Gaussian heuristic is loose in so small a rank. The minimum left undetermined in [11] is therefore available numerically, and the module reduction shortens the lifted vector by up to $6.24\times$.

Where the Gaussian heuristic stops holding. The third panel of Figure 2 shows $\lambda_1(\mathcal{D}_s)/\text{gh}(\mathcal{D}_s)$ at every second section rank, at one of the two moduli below the fatigue point where we measure 12 instances with $n = 64$. The heuristic is accurate up to $k \approx 0.8n$, after which the ratio decreases gradually and reaches 0.56 at $k = n$, close to $\Delta_{1,n} \sqrt{2\pi e/n}$; the decrease spans about a quarter of the range of k , in agreement with the width of Theorem 11, so there is no jump. Averaged over all instances and ranks the Gaussian heuristic misses the minimum by 19.2% and (2) by 8.6%, and from $k = 0.85n$ on, where the two differ, by 35–88% and 0–16%.

Attack comparison. Table II gives the cost measured at three ring degrees. The full-field reduction is the slowest wherever it terminates, by 17 to $3.5 \cdot 10^3$ in time and 6.4 to 34 in memory. Against the norm and the trace descent we gain up to $55\times$ and $47\times$ in time and $4.6\times$ and $4.7\times$ in memory, because their rules keep a relative degree our threshold already allows to double. Against the subring projection, the strongest earlier method, the gain reaches $5.3\times$ in time and $3.5\times$ in memory where the sharper rule drops one dimension level, and elsewhere the two agree. Memory does not always follow time, because a smaller dimension is sometimes traded against a larger block size. At $n = 512$ and $\log_2 q = 64$ the models predict $2^{34.0}$ operations for ours against $2^{36.3}$ for the subring projection and $2^{46.0}$ for the full field, and the last panel of Figure 2 shows that gap widening with the modulus, since the relative degree keeps rising while the full-field dimension stays at $2n$.

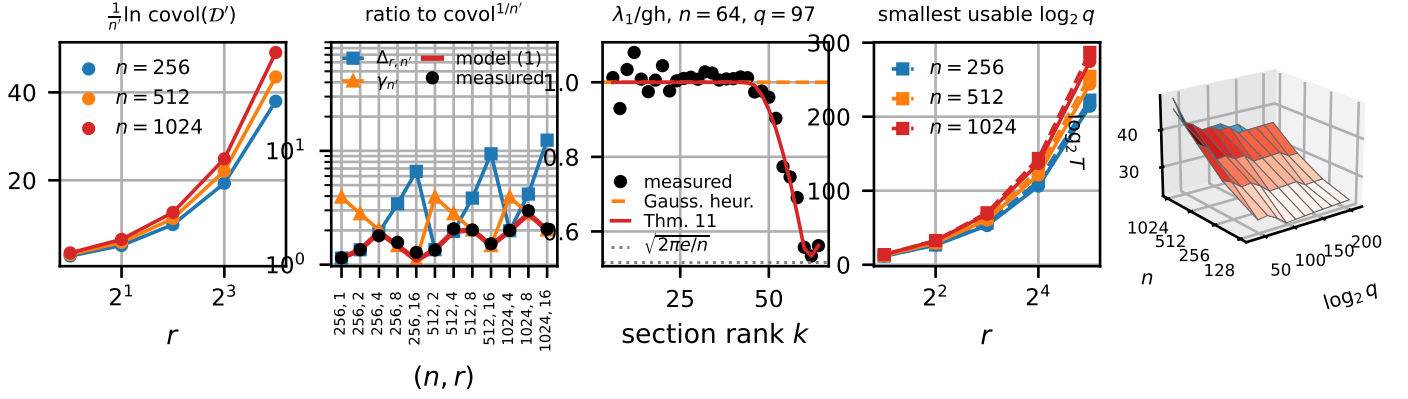


Fig. 2: Validation of Sections III and IV and predicted cost. First panel: $\frac{1}{n} \ln \text{covol}(\mathcal{D}')$ against the relative degree r , dots for measurements on ternary instances and lines for Theorem 5. Second: for each (n, r) the shape defect $\Delta_{r,n'}$, the constant $\gamma_{n'}$, their minimum, which is the model (1), and the measured $\lambda_1(\mathcal{D}')/\text{covol}(\mathcal{D}')^{1/n'}$, the gap between the two upper curves being what the module reduction gains. Third: the first minimum of a section over its Gaussian-heuristic value against the section rank k , dots averaged over six instances, the dashed line at 1 for [14] and the solid curve for (2). Fourth: the smallest modulus at which a relative degree is usable, with the module reduction of Algorithm 1 as solid lines and without it as dashed lines. Last: predicted logarithmic time against modulus and ring degree, the upper surface the best earlier attack and the lower one ours.

TABLE II: Measured cost of the five attacks on identical ternary instances, giving for each modulus the reduction dimension d the method chooses, its median time t in seconds, its median peak memory M in megabytes, and last the ratios of the best earlier time and memory to ours. A dash means the method was not run there and >bud that it exceeded its budget of ten minutes; the full-field reduction was run at every modulus for $n = 128$, three of which appear here, and at the cheapest one for $n = 256$, where it already exceeds that budget.

$\log_2 q$	Full field [13], [14]			Norm [11]			Trace [12]			Subring [13]			This work			gain	
	d	t	M	d	t	M	d	t	M	d	t	M	d	t	M	t	M
ring degree $n = 128$																	
24	256	6.917	21.41	128	0.854	5.31	128	0.855	4.83	96	0.396	3.33	96	0.395	3.33	1.0	1.0
56	256	37.909	25.58	32	0.045	0.91	64	0.252	1.70	32	0.043	1.11	32	0.045	0.86	0.9	1.1
112	256	57.173	26.98	16	0.016	0.84	16	0.016	0.80	16	0.017	1.19	16	0.016	0.80	1.0	1.0
ring degree $n = 256$																	
28	—	—	—	256	247.024	24.14	256	275.258	21.38	192	117.131	12.55	192	132.434	12.77	0.9	1.0
32	>bud	—	—	256	136.343	21.48	256	116.164	23.89	128	2.292	5.81	128	2.475	5.08	0.9	1.1
40	—	—	—	256	14.458	23.16	256	14.724	22.02	128	1.961	5.84	128	2.081	5.05	0.9	1.2
44	—	—	—	128	1.893	5.30	128	1.948	5.42	128	1.778	5.94	96	0.781	4.26	2.3	1.2
56	—	—	—	128	6.354	5.08	128	6.489	5.39	96	2.056	4.09	64	1.017	3.02	2.0	1.4
80	—	—	—	64	0.852	1.89	64	0.882	1.86	64	0.803	2.61	64	0.834	1.97	1.0	0.9
112	—	—	—	64	0.776	2.09	64	0.776	2.16	48	0.297	2.22	32	0.177	2.08	1.7	1.0
160	—	—	—	32	0.174	1.38	32	0.185	1.30	32	0.179	2.08	32	0.176	1.41	1.0	0.9
ring degree $n = 512$																	
56	—	—	—	256	593.201	21.19	256	628.350	4.81	192	104.902	6.75	192	104.530	6.75	1.0	0.7
64	—	—	—	256	373.680	5.59	256	579.975	13.45	192	167.740	21.68	128	31.697	24.05	5.3	0.2
80	—	—	—	128	19.926	5.84	256	240.662	5.12	128	19.008	8.88	128	20.578	3.84	0.9	1.3
120	—	—	—	128	17.042	6.22	128	17.086	6.63	96	5.796	8.33	96	5.677	8.20	1.0	0.8
128	—	—	—	128	17.289	4.46	128	17.022	6.52	96	5.463	8.18	64	2.995	2.30	1.8	1.9

Future Directions

Extending Theorem 5 to conductors with small odd prime factors would widen the range the predictor covers. A module reduction using the log-unit lattice of L would shorten the recovered vector further.

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VII. APPENDIX

A. Proof of Theorem 5

We first state the distribution of the embeddings of a Gaussian ring element. Let $\zeta = e^{i\pi/n}$ and let $\sigma_j(a) = a(\zeta^j)$ for odd j with $1 \leq j < 2n$ denote the n complex embeddings of K , which come in the $n/2$ conjugate pairs $\{\sigma_j, \sigma_{2n-j}\}$. Fix a set J of representatives, one from each pair.

Let $f = \sum_{k < n} f_k X^k$ with f_k independent and $N(0, \sigma^2)$. For $j \in J$ write $\sigma_j(f) = \Re_j + i \Im_j$ with $\Re_j = \sum_k f_k \cos(\pi j k / n)$ and $\Im_j = \sum_k f_k \sin(\pi j k / n)$. All these variables are jointly Gaussian and centered, so it suffices to compute second moments. Since j is odd and $1 \leq j < 2n$, the integer n does not

divide j , hence $\sum_{k < n} \cos(2\pi j k / n) = \Re \sum_{k < n} e^{2\pi i j k / n} = 0$ and likewise $\sum_{k < n} \sin(2\pi j k / n) = 0$. Therefore

$$\begin{aligned} \text{Var}(\Re_j) &= \sigma^2 \sum_k \cos^2\left(\frac{\pi j k}{n}\right) \\ &= \sigma^2 \left(\frac{n}{2} + \frac{1}{2} \sum_k \cos \frac{2\pi j k}{n} \right) = \frac{n\sigma^2}{2}, \end{aligned}$$

and the same computation gives $\text{Var}(\Im_j) = n\sigma^2/2$ and $\text{Cov}(\Re_j, \Im_j) = \frac{\sigma^2}{2} \sum_k \sin(2\pi j k / n) = 0$. For $j \neq j'$ in J we have $j \not\equiv \pm j' \pmod{2n}$, so all four covariances between $\{\Re_j, \Im_j\}$ and $\{\Re_{j'}, \Im_{j'}\}$ vanish by the product-to-sum identities and the same two vanishing sums. Being jointly Gaussian and uncorrelated, the $n/2$ pairs are independent, and each $\sigma_j(f)$ is a centered complex Gaussian with $\mathbb{E}|\sigma_j(f)|^2 = n\sigma^2$. Consequently $|\sigma_j(f)|^2 = n\sigma^2 a_j$ with a_j independent standard exponential variables, because the sum of the squares of two independent centered Gaussians of variance $v/2$ is v times a standard exponential.

We next identify which embeddings multiply together in the relative norm. The subgroup $H = \text{Gal}(K/L)$ consists of the σ_j with $j \equiv 1 \pmod{2n'}$, and the embeddings of K lying above a fixed embedding σ'_i of L , with i odd and $1 \leq i < 2n'$, are the σ_j with $j \equiv i \pmod{2n'}$; there are exactly r of them. If two of them were conjugate we would have $j' \equiv -j \pmod{2n}$ and $j' \equiv j \pmod{2n'}$, hence $2j \equiv 0 \pmod{2n'}$ and $n' \mid j$, which is impossible for odd j once $n' \geq 2$. So the r embeddings above σ'_i are pairwise non-conjugate, the classes above σ'_i and above its conjugate $\sigma'_{2n'-i}$ are disjoint, and the $n/2$ conjugate classes of K split into $n'/2$ groups of r classes, one group per conjugate class of L .

Fix a conjugate class of L and let $a_1, \dots, a_r$ and $b_1, \dots, b_r$ be the exponential variables attached to f and to g at the r classes of K above it. Then

$$|\sigma'(f')|^2 = \prod_{l \leq r} |\sigma_l(f)|^2 = (n\sigma^2)^r \prod_{l \leq r} a_l,$$

and similarly for g' , so $W_{\sigma'} = (n\sigma^2)^r \Psi$ with $\Psi = A + B$, $A = \prod_l a_l$ and $B = \prod_l b_l$. Distinct conjugate classes of L use disjoint groups of embeddings, so the corresponding Ψ are independent, and f and g are independent, so A and B are independent.

By Lemma 3 the covolume is $\prod_{\sigma'} W_{\sigma'}^{1/2}$ over all n' embeddings, and conjugate embeddings carry equal weights, so $\ln \text{covol}(\mathcal{D}') = \sum_{\text{classes}} \ln W$, a sum of $n'/2$ independent copies. It remains to compute $\mathbb{E}[\ln \Psi]$. Writing $\ln(A + B)$ in the two ways $\ln A + \ln(1 + B/A)$ and $\ln B + \ln(1 + A/B)$ and averaging gives

$$\ln(A + B) = \frac{1}{2} \ln(AB) + \frac{1}{2} \ln(2 + A/B + B/A).$$

Setting $D = \ln A - \ln B$ we have $A/B + B/A = 2 \cosh D$ and $2 + 2 \cosh D = 4 \cosh^2(D/2)$, so the second term equals $\ln(2 \cosh(D/2))$. For the first term, $\mathbb{E}[\ln a] = -\gamma$ for a standard exponential variable, so $\mathbb{E}[\frac{1}{2} \ln(AB)] = -r\gamma$. For the second term, if a and b are independent standard exponentials then $\Pr[\ln a - \ln b \leq x] = \Pr[a \leq be^x] = \mathbb{E}_b[1 - e^{-be^x}] =$

$e^x/(1+e^x)$, which is the standard logistic distribution function; hence D is a sum of r independent standard logistic variables, which is the variable D_r of the statement. Combining the two terms gives $\mathbb{E}[\ln \Psi] = \mu_r$ and therefore $\mathbb{E}[\ln \text{covol}(\mathcal{D}')] = \frac{n'}{2}(r \ln(n\sigma^2) + \mu_r)$.

For the variance, $\ln \text{covol}$ is a sum of $n'/2$ independent copies of $\ln W$, and $\ln W = r \ln(n\sigma^2) + \frac{1}{2} \ln(AB) + \ln(2 \cosh(D_r/2))$. The variance of $\frac{1}{2} \ln(AB)$ is $\frac{1}{4} \cdot 2r \cdot \pi^2/6$ because $\text{Var}(\ln a) = \pi^2/6$, and $\ln(2 \cosh(x/2)) - |x|/2 \in [0, \ln 2]$ bounds the second contribution by $\text{Var}(|D_r|)/4 + O(1) \leq r\pi^2/12 + O(1)$. Both are $O(r)$, so $\text{Var}(\ln \text{covol}(\mathcal{D}')) = O(n'r)$.

For the asymptotic form, D_r has mean 0 and variance $r\pi^2/3$, and $\ln(2 \cosh(x/2)) = |x|/2 + \ln(1 + e^{-|x|})$. The central limit theorem together with uniform integrability of $|D_r|/\sqrt{r}$ gives $\mathbb{E}[D_r] = \sqrt{2r\pi/3}(1 + o(1))$, hence $\mathbb{E}[|D_r|]/2 = \sqrt{\pi r/6}(1 + o(1))$, while the remaining term lies in $[0, \ln 2]$. This yields $\mu_r = -r\gamma + \sqrt{\pi r/6} + O(1)$. Finally, for $r = 1$ the variable $\Psi = a + b$ has the Gamma distribution with shape 2, whose logarithmic mean is the digamma value $\psi(2) = 1 - \gamma$, so $\mu_1 = 1 - \gamma$. $\square$

B. Proof of Theorem 8

Because $\langle f' \rangle$ and $\langle g' \rangle$ are coprime there exist $F', G' \in \mathcal{O}_L$ with $f'G' - g'F' = q$. Consider the $\mathcal{O}_L$ -module M generated by (f', g') and (F', G') . The identities

$$\begin{aligned} f'(F', G') - F'(f', g') &= (0, q), \\ g'(F', G') - G'(f', g') &= (-q, 0) \end{aligned}$$

show that M contains $q\mathcal{O}_L^2$, and $[f'^{-1}]_q(f', g') \equiv (1, h')$ modulo q shows that M contains a generating set of $\Lambda_q^{h'}$ modulo $q\mathcal{O}_L^2$; hence $\Lambda_q^{h'} \subseteq M$. The determinant over L of the pair $\{(f', g'), (F', G')\}$ equals $f'G' - g'F' = q$, which is also the determinant of $\{(1, h'), (0, q)\}$, so M and $\Lambda_q^{h'}$ have the same covolume and therefore coincide.

Let $V = \text{span}_{\mathbb{R}}(\mathcal{D}')$ and let π be the perpendicular projection onto $V^\perp$. For a power-of-two cyclotomic field the multiplication by X is a signed cyclic shift of coordinates, hence an isometry of the coefficient embedding, so V is stable under the action of $\mathcal{O}_L$ and so is $V^\perp$; consequently π commutes with that action and $\Lambda^\perp := \pi((F', G')\mathcal{O}_L) = w\mathcal{O}_L$ with $w = \pi(F', G')$, a rank-one $\mathcal{O}_L$ -module. Splitting the covolume of M along V and $V^\perp$ gives $\text{covol}(\mathcal{D}')\text{covol}(\Lambda^\perp) = \text{covol}(\Lambda_q^{h'}) = q^{n'}$.

We now apply the lower bound of Lemma 4 to $\Lambda^\perp$. Its proof only used that the lattice is the image of $\mathcal{O}_L$ under multiplication by a fixed vector, so it applies to any rank-one $\mathcal{O}_L$ -module: writing $\|vw\|^2 = \frac{1}{n'} \sum_{\sigma'} |\sigma'(v)|^2 W_{\sigma'}^\perp$ with $W_{\sigma'}^\perp = \sum_i |\sigma'(w_i)|^2$, the inequality between the arithmetic and the geometric mean and $|N_{L/\mathbb{Q}}(v)| \geq 1$ give $\|vw\| \geq (\prod_{\sigma'} W_{\sigma'}^\perp)^{1/(2n')} = \text{covol}(\Lambda^\perp)^{1/n'}$. Hence

$$\lambda_1(\Lambda^\perp) \geq \text{covol}(\Lambda^\perp)^{1/n'} = \frac{q}{\text{covol}(\mathcal{D}')^{1/n'}}.$$

Let $(x', y') \in \Lambda_q^{h'}$ satisfy $\|(x', y')\| < q/\text{covol}(\mathcal{D}')^{1/n'}$. Since $M = \Lambda_q^{h'}$, we may write $(x', y') = d + v(F', G')$ with $d \in \mathcal{D}'$ and $v \in \mathcal{O}_L$. Applying π gives $\pi(x', y') = vw \in \Lambda^\perp$, and the projection does not increase length, so $\|vw\| \leq \|(x', y')\| < \lambda_1(\Lambda^\perp)$. This forces $vw = 0$, hence $v = 0$ and $(x', y') = d \in \mathcal{D}'$.

For the comparison, the argument of [11] replaces $\text{covol}(\mathcal{D}')$ by the Hadamard bound $\|(f', g')\|^{n'}$, which yields the threshold $q/\|(f', g')\|$. By Definition 6 the two thresholds differ exactly by $\Delta_{r,n'} = \|(f', g')\|/\text{covol}(\mathcal{D}')^{1/n'}$, which is at least 1 by Corollary 7. $\square$

C. Proof of Theorem 11

Write m for the rank of $\mathcal{D}$ and k for the rank of the section $\mathcal{D}_s$. By Heuristic 10 the expected number of nonzero points of $\mathcal{D}_s$ inside a ball of radius R equals $\varrho_{\mathcal{D}}(R)V_k R^k / \text{covol}(\mathcal{D}_s)$. Equating this count to one and solving for R gives

$$\begin{aligned} R &= \left(\frac{\text{covol}(\mathcal{D}_s)}{V_k \varrho_{\mathcal{D}}(R)} \right)^{1/k} = \gamma_k \text{covol}(\mathcal{D}_s)^{1/k} \varrho_{\mathcal{D}}(R)^{-1/k} \\ &= \text{gh}(\mathcal{D}_s) \varrho_{\mathcal{D}}(R)^{-1/k}, \end{aligned}$$

which by Definition 9 is the second entry of the maximum in (2). The first entry is the inclusion $\mathcal{D}_s \subseteq \mathcal{D}$, which gives $\lambda_1(\mathcal{D}_s) \geq \lambda_1(\mathcal{D})$ unconditionally, and the maximum of the two is the value consistent with both constraints.

Assume now that $\mathcal{D}$ is a scaled rotation of $\mathbb{Z}^m$, so that after rescaling by $\text{covol}(\mathcal{D})^{1/m}$ we may count integer points. For $t > 0$ let p_t be the probability measure on $\mathbb{Z}$ with $p_t(z) \propto e^{-\pi t z^2}$ and let $\theta_3(t) = \sum_{z \in \mathbb{Z}} e^{-\pi t z^2}$ be its normalizing constant, so that the product measure $p_t^{\otimes m}$ has total mass $\theta_3(t)^m$ before normalization. For $S(z) = \|z\|^2$ we have the exact identity

$$\#\{z \in \mathbb{Z}^m : S(z) \leq \rho^2\} = \theta_3(t)^m \mathbb{E}_t[e^{\pi t S} \mathbf{1}_{S \leq \rho^2}].$$

Choose t so that $\mathbb{E}_t[S] = \rho^2$, which is possible because $t \mapsto \mathbb{E}_t[S] = m \mathbb{E}_t[z^2]$ is continuous and decreases from $+\infty$ to 0. Under $p_t^{\otimes m}$ the variable S is a sum of m independent integer-valued variables with mean ρ^2/m and variance $\text{Var}_t(z^2)$, so the local limit theorem for lattice-valued sums gives $\Pr_t[S = \rho^2 - u] = \frac{1}{s\sqrt{2\pi}}(1 + o(1))$ for $u = O(s)$, with $s^2 = m \text{Var}_t(z^2)$. Summing the geometric weights,

$$\begin{aligned} \mathbb{E}_t[e^{\pi t S} \mathbf{1}_{S \leq \rho^2}] &= e^{\pi t \rho^2} \sum_{u \geq 0} e^{-\pi t u} \Pr_t[S = \rho^2 - u] \\ &= \frac{e^{\pi t \rho^2} (1 + o(1))}{s\sqrt{2\pi} (1 - e^{-\pi t})}. \end{aligned}$$

Taking logarithms and dividing by m , the correction term is $O(\ln m/m)$, so with $\rho = c\sqrt{m}$ we obtain $\frac{1}{m} \ln \#\{z : \|z\| \leq c\sqrt{m}\} = \pi t c^2 + \ln \theta_3(t) + O(\ln m/m)$, and since the chosen t is exactly the stationary point of $t \mapsto \pi t c^2 + \ln \theta_3(t)$, the exponent equals $J(c) = \min_{t>0} \{\pi t c^2 + \ln \theta_3(t)\}$. The Gaussian heuristic count in the same ball has exponent $\frac{1}{m} \ln(V_m c^m m^{m/2}) = \ln(c\sqrt{2\pi e}) + O(\ln m/m)$, so $\Xi_{\mathcal{D}}(c) = J(c) - \ln(c\sqrt{2\pi e})$.

The function J is the infimum of a family of affine functions of c^2 , hence concave in c^2 and smooth wherever the minimizer is unique, which holds because $t \mapsto \pi t c^2 + \ln \theta_3(t)$ is convex

with a unique minimizer. Differentiating at the stationary point gives $J'(c) = 2\pi tc$, so $\Xi'(c) = 2\pi tc - 1/c$. The measure p_t is a discrete Gaussian on $\mathbb{Z}$ with parameter $s^2 = 1/t$, and the variance of a discrete Gaussian over $\mathbb{Z}$ is at most that of the continuous one, so $\mathbb{E}_t[z^2] \leq 1/(2\pi t)$, which together with $\mathbb{E}_t[z^2] = c^2$ gives $2\pi tc^2 \leq 1$ and hence $\Xi'(c) \leq 0$, with a decrease at each $c > 0$. So Ξ is smooth and decreasing.

For the two limits, first let c grow. The minimizing t then tends to 0, and the modular transformation $\theta_3(t) = t^{-1/2}\theta_3(1/t) = t^{-1/2}(1 + 2e^{-\pi/t} + O(e^{-4\pi/t}))$ turns the objective into $\pi tc^2 - \frac{1}{2} \ln t + 2e^{-\pi/t} + O(e^{-4\pi/t})$. Ignoring the exponentially small term, the minimizer is $t = 1/(2\pi c^2)$ with value $\frac{1}{2} \ln(2\pi ec^2) = \ln(c\sqrt{2\pi e})$; substituting the minimizer into the exponentially small term gives $\Xi(c) = 2e^{-2\pi^2 c^2}(1 + o(1))$. Second, let c tend to 0. The minimizing t then grows, $\theta_3(t) = 1 + 2e^{-\pi t} + O(e^{-4\pi t}) \rightarrow 1$, so $J(c) \rightarrow 0$ and $\Xi(c) = -\ln(c\sqrt{2\pi e}) + o(1)$.

Finally, the Gaussian heuristic is accurate for $\mathcal{D}_s$ within a factor $1 + \varepsilon$ exactly when $\varrho_{\mathcal{D}}^{1/k} \leq 1 + \varepsilon$, that is when $\frac{m}{k} \Xi(u) \leq \ln(1 + \varepsilon)$. Substituting the large- u form of Ξ gives $2e^{-2\pi^2 u^2} \leq \frac{k}{m} \ln(1 + \varepsilon)$, which is (3). Solving $\frac{m}{k} 2e^{-2\pi^2 u^2} = 1$ and $\frac{m}{k} 2e^{-2\pi^2 u^2} = 1/100$ and subtracting shows that the corresponding values of u^2 differ by $\ln(100)/(2\pi^2) = 0.2333$, independently of m and k , while the threshold itself satisfies $(u^*)^2 = \frac{1}{2\pi^2} \ln(2m/k) + O(1)$; the two are of the same order, so the ratio $\lambda_1(\mathcal{D}_s)/\text{gh}(\mathcal{D}_s)$ changes by a constant factor only over a range of u comparable to u^* itself. $\square$

D. Proof of Theorem 14

We account for the four stages separately. At level i of the descent the current element lies in a ring of degree $n/2^i$ and its coefficients are reduced modulo q , so one negacyclic product costs $O((n/2^i) \log(n/2^i) \log q \log \log q)$ bit operations with a fast integer product, and the extraction of the even part is linear. Summing over the $\log_2 r$ levels, and using that the degrees decrease geometrically, gives $O(n \log r \log q \log \log q)$ bit operations; the memory used is $O(n \log q)$ bits. The lift performs one product of the same shape in the full ring, which is inside the same bound.

The reduction stage first calls the L^2 algorithm on a basis of dimension d whose entries are bounded by q , at a cost of $O(d^4 \log q)$ bit operations and $O(d^2 \log q)$ bits of storage [25]. It then runs progressive tours of block reduction. At block size β one tour performs $d - \beta + 1$ calls to a shortest vector oracle in dimension β ; charging each call $2^{0.292\beta+16.4}$ operations and $2^{0.2075\beta+16.4}$ vectors of $O(\beta)$ words [21], [22], and using that the number of block sizes visited before halting is at most β while the cost of a call grows geometrically in β , the total is $O(d\beta 2^{0.292\beta+16.4})$ operations and $O(\beta 2^{0.2075\beta+16.4})$ bits, the latter dominating the basis storage as soon as β exceeds a small constant times $\log d$.

The module reduction operates on a lattice of rank n' generated by at most two blocks of n' vectors inside $\mathbb{Z}^{2n'}$, whose entries are bounded by q because they are products of a

lattice vector by a power of Y . One call to the L^2 algorithm in dimension $2n'$ therefore costs $O(n'^4 \log q)$ bit operations and $O(n'^2 \log q)$ bits. Adding the four contributions gives the stated bounds, and the reduction stage dominates the other three as soon as $d \geq n^{1/2}$, which holds for every configuration selected by the predictor. $\square$